Gravity on the 7+1 spacetime: symbols and their definitions

A dictionary of every symbol used for the curvature, the Einstein and Lovelock tensors, the generalized frame matrices, the canonical spin connection, the covariant Dirac equation of dirac16complex and the three cosmological epochs — sections 11–14 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation and its meaning. The symbols are grouped by topic in the order in which they are introduced; longer defining equations follow each table as displayed formulas. The last sections list symbols that look alike and the Mathematica names defined in sections 11–14 of 7+1.nb. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$. A prime is d/dx_0 for functions of the time x_0 (and d/dr for the functions $f(r)$, $h(r)$ of the static radial metric (2.4)).
- Index ranges: Greek $\mu, \nu, \lambda, \rho, \sigma, \alpha, \beta, \gamma, \delta \in \{0, \ldots, 7\}$ are coordinate indices; Latin capitals $A, B, C, D \in \{0, \ldots, 7\}$ are frame indices; $i, j \in \{1, 2, 3\}$ label ordinary space and $k, l \in \{4, \ldots, 7\}$ the auxiliary space (for the static radial metric (2.4), $k \in \{2, \ldots, 7\}$). Repeated indices are summed.
- The abstract indices of the xAct computation (section 4.3) are the letters a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y; they are unrelated to the index ranges above.
- Frame indices are raised and lowered with η, coordinate indices with $g_{\mu\nu}$.
- Signature: one time-like dimension (x_0, $g_{00} > 0$) and seven space-like dimensions ($x_1, \ldots, x_7$, negative squares). The xAct declaration $\{7, 1, 0\}$ counts them (seven space-like, one time-like, none degenerate), as in the reference notebook; the matrix $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ is a different object, simply the squares of the octonion basis ($e_0^2 = +1$, $e_i^2 = -1$, $\eta_{AB} = \mathrm{Re}(e_A e_B)$).
- Matrices: 1_n is the $n \times n$ identity, M^{T} the transpose, $[X, Y] = XY - YX$, $\{X, Y\} = XY + YX$.

- Mathematica lists are 1-based: $g_{\mu\nu}$ is `g[[μ+1, ν+1]]`, Γ_A is `Γ17[A]` (an argument, not a list index). Greek letters in names are named characters: `\[CapitalGamma]` is Γ, `\[Eta]` is η, `\[Alpha]` is α, `\[Beta]` is β, `\[Delta]` is δ, `\[CapitalLambda]` is Λ.
- Christoffel symbols are written $\{^{\lambda}_{\mu\nu}\}$, never Γ: in this project Γ always denotes a Dirac matrix.

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Spacetime, metrics and coordinates

symbol	Mathematica	definition	meaning
X; $x_0,\ldots,x_7$	`X = {x0, …, x7}`	$X = (x_0, x_1, \ldots, x_7)$	x_0 time (the octonion unit direction), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space
η_{AB}	`η17`	$\eta = \mathrm{diag}(+1,-1,\ldots,-1)$, $\eta_{AB} = \mathrm{Re}(e_A e_B)$	frame metric: simply the squares of the octonion basis, $e_0^2 = +1$ (time), $e_i^2 = -1$ (space)
η_{71}	`η71`	$-\eta = \mathrm{diag}(-1,+1,\ldots,+1)$	the same spacetime in the sign convention of 4+4.nb (space-like = $+1$): seven entries $+1$ (space-like), one -1 (time-like), as the xAct list $\{7,1,0\}$ counts them
e_A; xy; $\mathrm{Re}(x)$	`ev[A]`; `oMul[x, y]`; `oRe[x]`	octonion basis units $e_0 = 1, e_1, \ldots, e_7$, octonion product, real part (section 3 of 7+1.nb)	used in section 12.2 to check $\eta_{AB} = \mathrm{Re}(e_A e_B)$ and $\mathrm{Re}(xx) = x \cdot \eta \cdot x$
$g_{\mu\nu}$, $g^{\mu\nu}$	`cv["g"]`, `cv["ginv"]`	$g^{\mu\lambda}g_{\lambda\nu} = \delta^{\mu}_{\nu}$	metric and inverse metric
$\lvert g\rvert$	—	$\lvert\det g_{\mu\nu}\rvert$	volume factor $\sqrt{\lvert g\rvert}$
a, b	`a3[x0]`, `b4[x0]`	cosmological metric (2.1)	scale factors of ordinary and auxiliary space
H_a, H_b	`Ha`, `Hb`	$H_a = a'/a$, $H_b = b'/b$	Hubble rates
H_a', H_b'	`dHa`, `dHb`	derivatives of the Hubble rates	`toH` rewrites a', a'', b', b'' with them

symbol	Mathematica	definition	meaning
N	`nL[x0]`	$g_{00} = N^2$	lapse function (variational check)
n, a, b (inhomogeneous)	`nP`, `aP`, `bP`	polynomials (2.2)	inhomogeneous test metric
F	`fPP`	arbitrary function of $(x_0 - x_1, x_2, x_3)$	profile of the plane wave (2.3)
l_μ	`lNull`	$l_\mu = (1, -1, 0, \ldots, 0)$, $\eta^{\mu\nu} l_\mu l_\nu = 0$	null vector of the plane wave
r; f, h	`x1`; `fR[x1]`, `hR[x1]`	static radial metric (2.4)	radial coordinate and two arbitrary positive functions
C	`cConf[x0]`	$g = C(x_0)^2 \eta$	conformal factor
H, q_H	`H`, `qH`	$a = e^{H x_0}$, $b = e^{q_H H x_0}$	constant Hubble rate and ratio of the exponential vacuum solutions (4.9); H is also the inverse length of the field equations (4.7)

(2.1) Cosmological metric:

$$ds^2 = dx_0^2 - a(x_0)^2\big(dx_1^2 + dx_2^2 + dx_3^2\big) - b(x_0)^2\big(dx_4^2 + dx_5^2 + dx_6^2 + dx_7^2\big).$$

(2.2) Inhomogeneous metric (checked exactly at the point $x_0 = 1/2$, $x_4 = 1/3$, code `ptP`):

$$ds^2 = n^2 dx_0^2 - a^2 \sum_{i=1}^{3} dx_i^2 - b^2 \sum_{k=4}^{7} dx_k^2, \quad n = 1 + \tfrac{x_0 x_4}{5} + \tfrac{x_0^2}{11}, \quad a = 1 + \tfrac{x_0}{3} + \tfrac{x_4^2}{7}, \quad b = 2 - \tfrac{x_0 x_4}{4} + \tfrac{x_0^2}{9} + \tfrac{x_4}{13}.$$

(2.3) Plane gravitational wave (pp-wave):

$$ds^2 = dx_0^2 - dx_1^2 - \cdots - dx_7^2 + F(x_0 - x_1, x_2, x_3)\,(dx_0 - dx_1)^2.$$

(2.4) Static metric with a radial coordinate $r = x_1 > 0$:

$$ds^2 = f(r)\,dx_0^2 - h(r)\,dr^2 - r^2\big(dx_2^2 + \cdots + dx_7^2\big).$$

3. Curvature

symbol	Mathematica	definition	meaning
$\{^{\lambda}_{\mu\nu}\}$	`cv["chr"]`	(3.1)	Christoffel symbols; `chr[[λ+1, μ+1, ν+1]]`
$R^{\rho}{}_{\sigma\mu\nu}$	`cv["riem"]`	(3.2)	Riemann tensor (Landau–Lifshitz convention)
$R_{\rho\sigma\mu\nu}$	`riemLow[cv]`	$g_{\rho\lambda} R^{\lambda}{}_{\sigma\mu\nu}$	Riemann tensor, all indices down
$R^{\gamma\delta}{}_{\alpha\beta}$	`cv["rud"]`, `rudAssoc`	$g^{\delta\sigma} R^{\gamma}{}_{\sigma\alpha\beta}$	two indices up, two down; `rudAssoc` lists the non-zero entries with $\gamma < \delta$, $\alpha < \beta$
$R_{\sigma\nu}$	`cv["ric"]`	$R^{\rho}{}_{\sigma\rho\nu}$	Ricci tensor
R	`cv["R"]`	$g^{\sigma\nu} R_{\sigma\nu}$	scalar curvature
$G^{\mu}{}_{\nu}$	`cv["G"]`	$R^{\mu}{}_{\nu} - \frac{1}{2}\delta^{\mu}_{\nu} R$	Einstein tensor
K_{AB}	`secK[cv, A, B]`	$R^{AB}{}_{AB}$ (no sum)	sectional curvature of the cosmological metric
$\nabla_\mu T^{\mu}{}_{\nu}$	`divergenceMixed`	(3.3)	divergence of a mixed tensor
—	`curvature[g, simp]`	—	the curvature engine: returns all of the above for a metric matrix

$$\{^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}\, g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big), \tag{3.1}$$

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{^{\rho}_{\sigma\nu}\} - \partial_\nu\{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\}, \tag{3.2}$$

$$\nabla_\mu T^{\mu}{}_{\nu} = \partial_\mu T^{\mu}{}_{\nu} + \{^{\mu}_{\mu\lambda}\}T^{\lambda}{}_{\nu} - \{^{\lambda}_{\mu\nu}\}T^{\mu}{}_{\lambda}. \tag{3.3}$$

4. Lovelock tensors and field equations

4.1 Generalized Kronecker delta and Lovelock tensors

symbol	Mathematica	definition	meaning
$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n}$	`gKD[up, dn]`	(4.1)	generalized Kronecker delta (explicit index values)
$\delta^{\text{upper}}_{\text{lower}}$	`kδ[lower, upper]`	(4.1) with xAct `delta`	the same with abstract indices (xAct)
$E^{(k)\mu}{}_{\nu}$	`lovelockE[ra, k]`; `Lovelock1/2/3`	(4.2)	Lovelock tensor of order k; `Lovelock`$k = E^{(k)b}{}_{a}$
L_k	`lovelockL[ra, k]`	(4.3)	Lovelock density of order k
$E^{(0)}$	`lovelockE[ra, 0]`	$-\frac{1}{2}\delta^{\mu}_{\nu}$	cosmological term
L_2	`lanczosDiag[cv][[1]]`	$R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Gauss–Bonnet density
$H_{\mu\nu}$	`lanczosDiag[cv][[2]]`	(4.4)	Lanczos tensor ($= E^{(2)}$)
$\mathcal{U}$, $\mathcal{V}$	`pairSets`, `orderedSeqs`	sets of k disjoint pairs; ordered sequences of pairs	index pairs of the pair-set algorithm (4.5)
sgn	`Signature[up]` `Signature[dn]`	sign of the permutation upper → lower	
$E^{(k)}$ (cosmological)	`ECos[[k+1]]`, `LCos[[k+1]]`	closed forms in H_a, H_b, H'_a, H'_b	Lovelock tensors and densities of (2.1)

$$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\left[\delta^{\mu_i}_{\nu_j}\right]^{n}_{i,j=1}, \tag{4.1}$$

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \tag{4.2}$$

$$L_k = \frac{1}{2^k}\,\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \tag{4.3}$$

$$H_{\mu\nu} = 2\big(RR_{\mu\nu} - 2R_{\mu\alpha}R^{\alpha}{}_{\nu} - 2R^{\alpha\beta}R_{\mu\alpha\nu\beta} + R_{\mu}{}^{\alpha\beta\gamma}R_{\nu\alpha\beta\gamma}\big) - \tfrac{1}{2}g_{\mu\nu}L_2, \tag{4.4}$$

$$E^{(k)\mu}{}_{\nu} = -2^{k-1}k!\sum_{\mathcal{U}}\sum_{\mathcal{V}}\operatorname{sgn}(\mu\,\mathcal{U}\to\nu\,\mathcal{V})\prod_{i=1}^{k} R^{\mathcal{V}_i}{}_{\mathcal{U}_i}. \tag{4.5}$$

4.2 The xAct objects (the method of the reference notebook Einstein-Lovelock-7+1-Nash.nb)

symbol	Mathematica	definition	meaning
M_8	`M8`	`DefManifold[M8, 8, {a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y}]`	the 8-dimensional manifold with its abstract indices

symbol	Mathematica	definition	meaning
g	`g71`	`DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`	abstract metric; $\{7, 1, 0\}$: seven space-like dimensions, one time-like, none degenerate (the declaration of the reference notebook)
∇	`CD`	Levi-Civita derivative of `g71`	after `CD = LC[g71]`: the connection of a component metric
chart	`chartM8`	`DefChart[chartM8, M8, {0, …, 7}, {x0[], …, x7[]}, …]`	coordinate chart; `x0[]` … `x7[]` coordinate scalars
${\delta_a}^b$	`delta[-a, b]`	Kronecker delta of xAct	
${R_{ab}}^{cd}$ etc.	`RiemannCD`, `RicciCD`, `RicciScalarCD`, `EinsteinCD`	curvature of `CD` (abstract, or components after the assignments)	
—	`MyArrayComponents`	`MyArrayComponents[expr] := expr // ToBasis[chartM8] // ComponentArray // ToValues // ToValues // Simplify`	component array of an expression
—	`toXc`, `fromXc`	$x_\mu \to$ `xμ[]` and back	between this notebook's coordinates and the chart scalars
—	`xActLovelock[gm]`	`g71 = CTensor[gm, {-chartM8, -chartM8}]`; `MetricCompute`; `CD = LC[g71]`; `RiemannCD = Riemann[CD]`, …; `MyArrayComponents` of `Lovelock1/2/3`, `EinsteinCD[-a, b]`, transposed	returns $E^{(1)}, E^{(2)}, E^{(3)}, G$ as matrices (row μ, column ν)
—	`xlCos`, `xlPP`, `xlExp`	`xActLovelock` of (2.1), (2.3) and of $a = e^{Hx_0}$, $b = e^{q_H H x_0}$	

4.3 Couplings and field equations

symbol	Mathematica	definition	meaning
κ, G_8	`kap`	$\kappa = 8\pi G_8$	gravitational coupling; Newton constant in 8 dimensions
α_k	`alf[k]`	(4.6)	Lovelock couplings, $\alpha_1 = 1$, α_2 (length^2), α_3 (length^4)
$\bar\Lambda$	`lamb`	$\alpha_0 = 2\bar\Lambda$	cosmological constant (length^{-2}); in the field equations of the three epochs (section 7) it is written Λ
w_1, w_2, w_3	`w1`, `w2`, `w3`	(4.7)	pure-number couplings of the reference notebook
Λ	`\[CapitalLambda]`	(4.7)	pure-number cosmological constant of the reference notebook
H	`H`	(4.7)	fundamental inverse length
${T^\mu}_\nu$	—	$\delta S_{\text{matter}} = \frac{1}{2}\int \sqrt{\lvert g\rvert} T_{\mu\nu}\delta g^{\mu\nu}$	energy–momentum tensor
ε, p_a, p_b	—	$T = \operatorname{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, \ldots, -p_b)$	energy density and pressures

$$S = -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\sum_{k=0}^{3}\alpha_k L_k + S_{\text{matter}}, \qquad \sum_{k=0}^{3}\alpha_k {E^{(k)\mu}}_\nu = \kappa {T^\mu}_\nu, \tag{4.6}$$

$$0 = -\Lambda\,\delta^{\mu}_{\nu} + H^{-2} w_1\, E^{(1)\mu}{}_{\nu} + H^{-4} w_2\, E^{(2)\mu}{}_{\nu} + H^{-6} w_3\, E^{(3)\mu}{}_{\nu}, \tag{4.7}$$

$$\alpha_2 = \frac{w_2}{w_1 H^2}, \quad \alpha_3 = \frac{w_3}{w_1 H^4}, \quad \bar{\Lambda} = \frac{\Lambda H^2}{w_1} \quad (T=0), \tag{4.8}$$

$$a = e^{H x_0}, \quad b = e^{q_H H x_0}: \quad H_a = H,\ H_b = q_H H,\ H'_a = H'_b = 0; \tag{4.9}$$

$$q_H = -1: \quad w_2 = \tfrac{\Lambda}{4} - \tfrac{w_1}{2},\ \ w_3 = \tfrac{w_1}{24} - \tfrac{\Lambda}{36}.$$

5. Frames, spin connection, spinor curvature

symbol	Mathematica	definition	meaning
$e^{A}{}_{\mu}$	`fd["e"]`	$g_{\mu\nu} = \eta_{AB} e^{A}{}_{\mu} e^{B}{}_{\nu}$	frame field (vielbein); `e[[A+1, μ+1]]`
$e_{A}{}^{\mu}$	`fd["ei"]`	$e_{A}{}^{\mu} e^{A}{}_{\nu} = \delta^{\mu}_{\nu}$	inverse frame; `ei[[μ+1, A+1]]`
Γ_A	`Γ17[A]`	$\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$	real 16×16 Dirac matrices, $\{\Gamma_A, \Gamma_B\} = 2\eta_{AB} 1_{16}$
Γ^A	`Γ17up[A]`	$\eta^{AA}\Gamma_A$ (no sum)	
$\Gamma_\mu(X)$	`fd["Gam"]`	$e^{A}{}_{\mu}\Gamma_A$	generalized frame matrices
$\Gamma^\mu(X)$	`fd["GamUp"]`	$e_{A}{}^{\mu}\Gamma^A$	
$\omega_{\mu}{}^{A}{}_{B}$	`fd["om"]`	(5.1)	canonical spin connection
$\omega_{\mu AB}$	`fd["omL"]`	$\eta_{AC}\omega_{\mu}{}^{C}{}_{B}$	antisymmetric in A, B
Ω_μ	`fd["Om"]`	$\frac{1}{4}\omega_{\mu AB}\Gamma^A\Gamma^B$	spinor connection (real 16×16)
$\mathcal{F}_{\mu\nu}$	`fd["F"]`	(5.2)	spinor curvature
$R_{AB\mu\nu}$	`rABmn`	$\eta_{AC} e^{C}{}_{\rho} R^{\rho}{}_{\sigma\mu\nu} e_{B}{}^{\sigma}$	Riemann tensor with two frame indices
$C^{A}{}_{BC}$	`structureC`	(5.3)	structure coefficients of the frame
$\omega_{C,AB}$	`omegaFrame`	$e_{C}{}^{\mu}\omega_{\mu AB}$	spin connection with a frame form index
1_{16}, 0_{16}	`ID16`, `Zero16`		identity and zero 16×16 matrices
—	`frameData`, `frameChecks`, `cartanEq`, `structureFormula`		compute and check the frame data of a metric

$$\omega_{\mu}{}^{A}{}_{B} = e^{A}{}_{\nu}\big(\partial_\mu e_{B}{}^{\nu} + \{{}^{\nu}_{\mu\lambda}\} e_{B}{}^{\lambda}\big), \tag{5.1}$$

$$\mathcal{F}_{\mu\nu} = \partial_\mu \Omega_\nu - \partial_\nu \Omega_\mu + [\Omega_\mu, \Omega_\nu] = \tfrac{1}{4} R_{AB\mu\nu}\Gamma^A\Gamma^B, \tag{5.2}$$

$$C^{A}{}_{BC} = e_{B}{}^{\mu} e_{C}{}^{\nu}\big(\partial_\mu e^{A}{}_{\nu} - \partial_\nu e^{A}{}_{\mu}\big), \qquad \omega_{C,AB} = -\tfrac{1}{2}\big(C_{ACB} + C_{BAC} - C_{CBA}\big). \tag{5.3}$$

The total covariant derivative and the vielbein postulate:

$$\nabla_\mu \Gamma_\nu = \partial_\mu \Gamma_\nu - \{{}^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0, \qquad \partial_\mu e^{A}{}_{\nu} - \{{}^{\lambda}_{\mu\nu}\} e^{A}{}_{\lambda} + \omega_{\mu}{}^{A}{}_{B} e^{B}{}_{\nu} = 0.$$

6. The covariant Dirac equation of dirac16complex

symbol	Mathematica	definition	meaning
ψ	`psiF`	16 functions of (x_0, x_1, x_5) in the checks	the field dirac16complex
$\bar{\psi}$	`psibF`	$\psi^\dagger\Gamma_0$ (independent functions in the checks)	Dirac adjoint
D_μ	`DpsiCos`, `DpsibCos`	(6.1)	spinor covariant derivative
m	`mD`		mass of dirac16complex
J^μ	—	$\bar{\psi}\Gamma^\mu\psi$	conserved current
χ	`chiF`	$\chi = (a^3b^4)^{1/2}\psi$	rescaled field (cosmological metric)
$c(x_0)$	`cModes`	$\chi = e^{i(\vec{p}\cdot\vec{x}+\vec{q}\cdot\vec{y})}c(x_0)$	mode function (16 components)
$\vec{p}$, $\vec{q}$	`pmom[i]`, `qmom[k]`	$\vec{p} = (p_1, p_2, p_3)$, $\vec{q} = (q_4, \ldots, q_7)$	comoving momenta along ordinary and auxiliary space
α_j, β	`αD[j]`, `βD`	$\alpha_j = \Gamma^0\Gamma^j$, $\beta = \Gamma_0$	Hamiltonian Dirac matrices
$h(x_0)$	`hCos`	(6.2)	mode Hamiltonian
A_μ, q, α	`aGauge[m]`, `qG`, `alGauge`	(6.3)	abelian gauge field, charge, gauge function
χ (radial)	`chiR`	$\psi = f^{-1/4}r^{-3}\chi$	rescaled field (static radial metric)

$$D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi, \qquad D_\mu\bar{\psi} = \partial_\mu\bar{\psi} - \bar{\psi}\,\Omega_\mu, \qquad (i\Gamma^\mu D_\mu - m)\psi = 0, \tag{6.1}$$

$$h(x_0) = \sum_{i=1}^{3}\alpha_i\frac{p_i}{a} + \sum_{k=4}^{7}\alpha_k\frac{q_k}{b} + \beta m, \qquad i\frac{dc}{dx_0} = h\,c, \tag{6.2}$$

$$D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi - iqA_\mu\psi, \qquad \psi \to e^{iq\alpha}\psi, \quad A_\mu \to A_\mu + \partial_\mu\alpha. \tag{6.3}$$

7. The three epochs

symbol	Mathematica	definition	meaning
p, q	`pK`, `qK`	$a = x_0^p$, $b = x_0^q$	Kasner exponents
p_E, q_E	`pKE`, `qKE`	$p = (1+2\sqrt{2})/7$, $q = (2-3\sqrt{2})/14$	Einstein–Kasner branch (expanding ordinary space, contracting auxiliary space)
K, λ	—	$K = \lvert\vec{p}\rvert\lambda^{1-p}$, $\lambda = (\lvert\vec{q}\rvert/\lvert\vec{p}\rvert)^{1/(q-p)}$	scaling variable of massless Kasner modes
$\mathcal{A}$, $\mathcal{B}$, $\mathcal{Z}$	`Aop`, `Bop2`, `Zc`	$\mathcal{A} = \vec{n}_a\cdot\vec{\alpha}$, $\mathcal{B} = \vec{n}_b\cdot\vec{\alpha}$, $\mathcal{Z} = \mathcal{A}\mathcal{B}\beta$	$\mathcal{Z}^2 = -1_{16}$: conserved $\pm i$
$\sigma_{1,2,3}$	`pauli`	Pauli matrices	2×2 reduction of the mode equation
$\lvert\alpha\rvert^2$, $\lvert\beta\rvert^2$	`bogoliubov[h, t0, t1, n]`	returns $\{\lvert\beta\rvert^2, \lvert\alpha\rvert^2\}$	Bogoliubov coefficients: pair-creation probability
b_0, $V_\perp$, V_c	—	$V_\perp = b_0^4 V_c$	static auxiliary scale, its proper and coordinate volume
G_4	—	$G_4 = G_8/V_\perp$	four-dimensional Newton constant
M	—	$\sqrt{m^2 + \lvert\vec{q}\rvert^2/b_0^2}$	Kaluza–Klein mass
H (de Sitter)	`hdS`	$a = e^{Hx_0}$	de Sitter expansion rate

symbol	Mathematica	definition	meaning
T_{GH}	—	$H/(2\pi)$	Gibbons–Hawking temperature

$$|\beta|^2 = \frac{1}{e^{2\pi M/H}+1} \quad \text{(de Sitter)}, \qquad |\beta|^2 = e^{-\pi\kappa_{\mathrm{LZ}}^2} \quad \text{(Landau–Zener benchmark, } i\dot{c} = (\kappa_{\mathrm{LZ}}\sigma_1 + t\sigma_3)c\text{)}.$$

8. Symbols that look alike

symbol	not to be confused with
Γ_A, Γ_μ (Dirac matrices)	$\{^{\lambda}_{\mu\nu}\}$ (Christoffel symbols; never written Γ here)
H (inverse length of (4.7); de Sitter rate)	H_a, H_b (Hubble rates), $H_{\mu\nu}$ (Lanczos tensor)
Λ (pure number, (4.7))	$\bar{\Lambda}$ (cosmological constant with dimension length^{-2})
α_k (Lovelock couplings)	α_j (Dirac matrices $\Gamma^0\Gamma^j$), α (gauge function), $\lvert\alpha\rvert^2$ (Bogoliubov)
β (Dirac matrix Γ_0)	$\lvert\beta\rvert^2$ (Bogoliubov coefficient)
q (charge; Kasner exponent of b; xAct index)	q_H (ratio of Hubble rates, code `qH`), $\vec{q}$ (auxiliary momentum)
w_1, w_2, w_3 (Lovelock couplings)	W^1, W^2, W^3 (electroweak gauge fields, code `wW1`, `wW2`, `wW3`, in sections 17–19 of 7+1.nb)
K (Kasner scaling variable)	K_{AB} (sectional curvature)
$\kappa = 8\pi G_8$	κ_{LZ} (Landau–Zener coupling)
`Lovelock1` $= E^{(1)b}{}_a$ (xAct)	`lovelockE[ra, 1]` $= E^{(1)\mu}{}_\nu$ (pair-set sum): the same tensor, row/column order transposed by `xActLovelock`
$\{7, 1, 0\}$ (xAct signature: seven space-like dimensions, one time-like)	$\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ (simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$)